

Kai Hei (Clement) Chu

📍 Vancouver, Toronto, Hong Kong

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Summary

A highly analytical, research-oriented undergraduate student, pursuing a Bachelor of Science in Mathematics and Its Applications specialist (Probability and Statistics). Passionate about Bayesian and probability theory. Fluent in Cantonese, Mandarin, and English, intermediate proficiency in Japanese. Proficient in Python, Excel and R. Expanding skillsets in SQL, TensorFlow, Keras packages.

Education

Sentinel Secondary School

Sept 2021 – June 2023

University of Toronto

Sept 2023 – (Expected)

BSc in Mathematics and Its Applications Specialist in Probability and Statistics,

April 2027

Minor in Statistics

- **Selected Coursework (*In Progress; selected core and projected advanced courses*):** Real Analysis I & II, Complex Variables, Linear Algebra I & II, Topology, Multivariable Calculus, ODE, Abstract Algebra, Probability theory, Stochastic processes(*), Theory of Statistics, Data analysis, Regression theory, Data Structures, Time Series (*), Mathematical Statistics I & II, Statistical Inference (*), Sampling theory, Spatial Data analysis (*), Methods for Multivariate Data (*), Statistical Computation (*)

()*: Graduate cross-listed courses.

Experience

Co-founder of Health and Nutrition club

Sentinel Secondary School

- Aimed to help students with mental health and stress through using facts and theories from food and nutrition.

Mentor for Science and Mathematics

Sentinel Secondary School

- Mentored students in science and mathematics. Developed valuable communication skills and problem-solving skills.

Volunteered in West Vancouver Senior Activity's Centre, West Vancouver

Vancouver

Memorial Library

- Volunteered for over 100 hours, demonstrating perseverance and significant contribution to community.
- Assisted children in library, developed valuable communication and explained instructions and activities in accessible ways.
- Assisted seniors in the different positions of the line of operation. Developed interpersonal communication skills, time management skills, and problem solving skills.

**Taiwan Republic of China Student Association (ROCSAUT)- Events Executive
(Executive team)**

*University of Toronto,
2024-Present*

- To be updated.

Chemistry Tutor

Self-employed, 2023-Present

- Provided tutoring for high school students in IB Chemistry Higher Level and British Columbia curriculums. Designed own teaching notes and problem sets for concept reinforcement.

PUM Math Tutor (Year long)

*University of Toronto,
2024-2025*

- Volunteered to provide one-to-one assistance to students struggling with high school mathematics, which helped boost student confidence and build an inclusive learning environment.

Volunteer

Staff of UofT Kids' Passport Event (Mathematics), Host assistance

University of Toronto

- Developed valuable communication skills by teaching children of various age about challenging mathematical concepts like Tower of Hanoi, making it more accessible for them.

Grader, COMC (Canadian Open Mathematics Competition), 2025

University of Toronto

- Assisted in grading the COMC papers in 2025, a national-level mathematics competition for IMO Canadian team qualifications.

Research Experience, Projects

Published Works

Research Project: *Preservation of Baire Spaces under Homeomorphisms*

University of Toronto, 2025

- An expository paper on point-set topology, accepted to University of Toronto Undergraduate Research Journal (TRUNK), First Edition. Click [here](#) (select TRUNK 2024-2025) to access the publication.

Ongoing Projects/Papers

Research Project: *Regularizing Boundary Degeneration of the MLE via Dirichlet Priors: A MAP-Based Perspective on the Probability Simplex*

University of Toronto, 2025

- In progress. Investigates the failure of the MLE on the boundary of the probability simplex, where it yields infinite KL divergence. Proposes a measure-theoretic regularization via MAP estimation under Dirichlet priors, with topological justifications. Received informal guidance from PhD candidate Fred Haochen Song (University of Toronto, Department of Statistical Sciences).

Research Project: *Analysis of Robustness of Linear Estimators via ℓ^p and Dual Norm Lipschitz Bounds*

University of Toronto, 2025

- An independent expository paper on an alternative perspective of measuring robustness of linear estimators (or linear functionals in finite dimensional vector spaces) by using techniques like Lipschitz continuity and norms. Received informal guidance from PhD candidate Sin Hang Jason Yeung (University of Toronto, Department of Mathematics). A copy of the draft is available [here](#).

Research-Related Activities

Contributor, editor of Openmath

- Openmath is an open source math website for self-learning mathematical topics. Content includes probability, machine learning, and other advanced math topics. Contributed to the proof writing part of the website using LaTeX and HTML. Enhanced proof-writing skills, delivered rigorous proofs accessible for different people from different math backgrounds, developed proficiency in HTML and LaTeX skills.

Directed Reading Course: Spectral Graph Theory

University of Toronto, 2025

- Selected for a program organized by MGSA (Mathematics Graduate Students Association), at the University of Toronto for advanced mathematical studies. Studied Spectral Graph Theory and its applications under the supervision of Dr. Narmada Varadarajan. The powerpoint for the presentation is available [here](#) [🔗](#).
- *Topics include:* Eigenvalues of graphs, Perron-Frobenius theorem, (n,d,c) -expanders, Expander-mixing lemma, Hoffman-Delsarte bound, Rayleigh quotients, Pseudorandom graphs, Laplacian spectrum with applications to graph drawings

Directed Reading Course: Topics in Causal Inference (*Ongoing*)

University of Toronto, Winter 2026

Advisor: Arian Hashemzadeh (PhD Student in Statistics, University of Toronto)

Awards

Dean's List Scholar

University of Toronto, 2024-2025

UBC Outstanding International Student Award

Awarded a total of 10000 CAD in scholarship. Declined.

University of Waterloo, President's Scholarship of Distinction from Waterloo

Awarded a scholarship valued at 5000 CAD. Declined.

PLKCKY- IGCSE public exam Scholarship

Awarded 2000 HKD, received 7A* (Highest grade) out of 10 subjects.

ABRSM- Piano Grade 7 (Merit)

Technologies

Languages: Python, R, Excel

Typesetting: L^AT_EX

Packages: ggplot2, dplyr

Libraries: matplotlib, seaborn, numpy, pandas

Interests

Research Interests: Bayesian inference and regularization, topology and its applications on statistical models, information theory

Currently developing skillsets in: Differential geometry (self-learning), information theory (self-learning), `TensorFlow`, `Keras`